Kick-off Diagnostic · L6-5 — Properties of Logarithms

Grade 11

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).
The one idea: A product inside a logarithm becomes a sum, a quotient becomes a difference, and an exponent comes out front; only a product or quotient splits.
The mistake most students make: Splitting a sum: $\log(M+N) = \log M + \log N$.
Readiness items 3, 4, 5, 7, 8 test: exponent laws, values of $\log_b x$ (Lesson 6-3), graph shifts of $\log_2 x$ (Lesson 6-4).
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All six stations, including the checks
B. I watched and read but skipped the checks
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, $\log_b(MN)$ becomes…

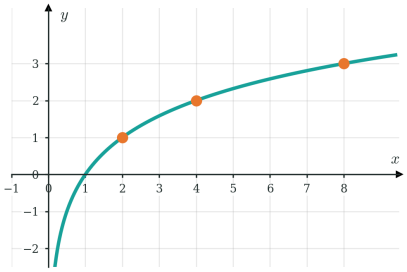
✓ **A.** $\log_b M + \log_b N$
B. $\log_b M \cdot \log_b N$
C. $\log_b M - \log_b N$
D. $\dfrac{\log_b M}{\log_b N}$

Signal. Wrong → did not watch station 1. Picking $\log_b M \cdot \log_b N$ (B) = copied the product sign.

3 FILL IN THE BLANK · 30 s — READINESS

Use the graph of $y = \log_2 x$ to find $\log_2 4 + \log_2 8 =$ _____

Accept: 5



Signal. 3 or 6 → misread the graph; 12 → added the inputs. Log values (Lesson 6-3) need re-teaching.

4 CHECKBOX · 30 s — READINESS

Select the **two** true statements.

✓ **A.** $2^3 \cdot 2^4 = 2^7$
B. $2^3 \cdot 2^4 = 4^7$
✓ **C.** $(2^3)^2 = 2^6$
D. $2^5 \div 2^2 = 2^{2.5}$

Signal. Choosing 4^7 or $2^{2.5}$ → exponent laws are the gap; the log properties are those laws in disguise.

5 MATCH · 30 s — READINESS

Match each logarithm with its value.

$\log_2 8$ → 3
$\log_3 9$ → 2 (shown shuffled in Wayground)
$\log_5 1$ → 0

Signal. Any miss → the definition of a logarithm (Lesson 6-3) is shaky. $\log_5 1 = 1$ is the common slip.

6 MULTIPLE CHOICE · 30 s **PRE-SESSION CHECK**

Which mistake did the pre-session warn most students make?

✓ **A. Splitting $\log(M+N)$ into $\log M + \log N$**

B. Writing $\log \dfrac{M}{N}$ as $\log M \div \log N$

C. Writing $\log M^k$ as $(\log M)^k$

D. Writing $\log_b b$ as b

Signal. Wrong → skimmed the key-idea slide; expect this error on the day.

7 REORDER · 30 s **READINESS**

Put in order, smallest to largest.

Correct order: $\log_2 \dfrac{1}{4} \rightarrow \log_2 1 \rightarrow \log_2 4 \rightarrow \log_2 32$

Signal. Wrong → negative and zero logs are not secure (-2, 0, 2, 5).

8 DROPDOWN · 30 s **READINESS**

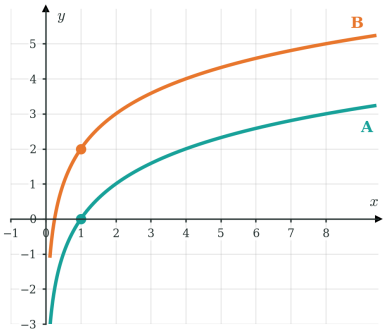
Curve A is $y = \log_2 x$. Curve B is A shifted up 2. The equation of B is...

✓ **A. $y = \log_2 x + 2$**

B. $y = \log_2(x+2)$

C. $y = \log_2(x-2)$

D. $y = \log_2 x - 2$



Signal. $\log_2(x+2)$ = moved the shift inside, the same mix-up the properties will clear up.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach exponent laws and log values with me. **3–4** → straight to the properties; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.